

Spin part 2

Let $|s\rangle$ be a spin with polar angle θ and azimuth angle ϕ .

$$|s\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) e^{i\phi} \end{pmatrix}$$

Spin measurement probabilities are the transition probabilities from $|s\rangle$ to an eigenstate.

The eigenstates are given by the eigenvectors

$$\begin{aligned} |x_+\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} & |y_+\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} & |z_+\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |x_-\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} & |y_-\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} & |z_-\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

Hence in the x direction

$$\begin{aligned} \text{Pr}(+) &= |\langle x_+ | s \rangle|^2 = \frac{1}{2} + \frac{1}{2} \sin \theta \cos \phi \\ \text{Pr}(-) &= |\langle x_- | s \rangle|^2 = \frac{1}{2} - \frac{1}{2} \sin \theta \cos \phi \end{aligned}$$

In the y direction

$$\begin{aligned} \text{Pr}(+) &= |\langle y_+ | s \rangle|^2 = \frac{1}{2} + \frac{1}{2} \sin \theta \sin \phi \\ \text{Pr}(-) &= |\langle y_- | s \rangle|^2 = \frac{1}{2} - \frac{1}{2} \sin \theta \sin \phi \end{aligned}$$

In the z direction

$$\begin{aligned} \text{Pr}(+) &= |\langle z_+ | s \rangle|^2 = \frac{1}{2} + \frac{1}{2} \cos \theta \\ \text{Pr}(-) &= |\langle z_- | s \rangle|^2 = \frac{1}{2} - \frac{1}{2} \cos \theta \end{aligned}$$

For each direction we have $\text{Pr}(+) + \text{Pr}(-) = 1$ as required by total probability.

Eigenmath script